By applying the reduction formula III iteratively, we get the following results (stated without proof):

$\int_0^1 x^m\ (1-x)^n dx\ =\ \dfrac{m!\times n!}{(m+n+1)!}$, where m and n are positive integers.

Example 9.42

Evaluate $\int_0^1 x^3\left(1-x\right)^4 dx$.

Solution

$$\int_0^1 x^m\left(1-x\right)^n dx\ =\ \frac{m!\times n!}{(m+n+1)!}.$$

$$\therefore \int_0^1 x^3\left(1-x\right)^4 dx\ =\ \frac{3!\times 4!}{(3+4+1)!}=\frac{3!\times 4!}{8!}=\frac{3\times 2\times 1\times 4\times 3\times 2\times 1}{8\times 7\times 6\times 5\times 4\times 3\times 2\times 1}=\frac{1}{280}.$$

EXERCISE 9.6

1. Evaluate the following:

(i) $\int_0^{\frac{\pi}{2}}\sin^{10} x\ dx$ (ii) $\int_0^{\frac{\pi}{2}}\cos^7 x\ dx$ (iii) $\int_0^{\frac{\pi}{4}}\sin^6 2x\ dx$ (iv) $\int_0^{\frac{\pi}{6}}\sin^5 3x\ dx$

(v) $\int_0^{\frac{\pi}{2}}\sin^2 x\cos^4 x\ dx$ (vi) $\int_0^{2\pi}\sin^7\frac{x}{4}\ dx$ (vii) $\int_0^{\frac{\pi}{2}}\sin^3\theta\cos^5\theta\, d\theta$ (viii) $\int_0^1 x^2(1-x)^3\ dx$

9.7 Gamma Integral

In this section, we study about a special improper integral of the form $\int_0^\infty e^{-x}x^{n-1}dx$, where n is a positive integer. Here, we have

$$e^\infty\ =\ \lim_{x\to\infty}e^x=\infty \text{ and } e^{-\infty}=\lim_{x\to\infty}e^{-x}=\frac{1}{\lim\limits_{x\to\infty}e^x}=\frac{1}{\infty}=0$$

By L'Hôpital's rule, for every positive integer m, we get,

$$\lim_{x\to\infty}x^m e^{-x}\ =\ \lim_{x\to\infty}\frac{x^m}{e^x}=\lim_{x\to\infty}\frac{m!}{e^x}=0.$$

Example 9.43

Prove that $\int_0^\infty e^{-x}x^n dx=n!$, where n is a positive integer.

Solution

Applying integration by parts, we get

$$\int_0^\infty e^{-x}x^n dx=\left[x^n\left(-e^{-x}\right)\right]_0^\infty-\int_0^\infty\left(-e^{-x}\right)\left(nx^{n-1}\right)dx=n\int_0^\infty e^{-x}x^{n-1}dx.$$

Let $I_n\ =\ \int_0^\infty e^{-x}x^n dx$. Then, $I_n=nI_{n-1}$.

So, we get $I_n\ =\ n(n-1)I_{n-2}$.

Proceeding in this way, we get ultimately,

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